

test-time operator A

operator-aware support stage

from y, A, λ

ISTA _{T}
iterate $x^{(T)}$

Coordinate features ϕ_j

- $|x_j^{(T)}|$ iterate
- $A_{:,j}^\top (y - Ax^{(T)})$ residual
- $\|A_{:,j}\|_2$ column norm
- $(A^\top A)_{jj}$ Gram diag.
- $\max_{i \neq j} |A_{:,i}^\top A_{:,j}|$ coherence

Coord-wise
classifier

$$\hat{p}_j = \sigma(w^\top \phi_j)$$

Estimate $\hat{x}$

Support-LS

$$\hat{x}_{\hat{S}} = A_{\hat{S}}^+ y$$

Top- k
support $\hat{S}$